

NLP — 10-Page Master Quick Revision

High-Yield Formula Sheet, Trellis Trace Matrices & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Natural Language Processing (CS24351)

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The Complete 53–Topic Syllabus Master Checklist

53 / 53 Syllabus Topics Complete

Module	Topic #	Topic Name	Status
M1: Intro & Preprocessing (12)	1	Introduction to NLP (Scientific vs. Engineering Goals) [UPLOADED PYQ]	✓ Complete
	2	Real-World Applications of NLP [UPLOADED PYQ]	✓ Complete
	3	Morphology (Morphemes, Free vs. Bound, Inflectional vs. Derivational)	✓ Complete
	4	Syntax (Phrase Structure & Grammatical Trees)	✓ Complete
	5	Semantics (Literal Meaning & Semantic Roles)	✓ Complete
	6	Pragmatics (Contextual Meaning & S-S-P Paradigm)	✓ Complete
	7	Tokenization (Word, Sentence, Subword BPE & Character)	✓ Complete
	8	Lemmatization (Base Lemma vs. Porter Stemmer) [UPLOADED PYQ]	✓ Complete
	9	Stop-Word Removal & Information Retrieval Tradeoffs	✓ Complete
	10	Normalization (Case Folding, Unicode, Contractions) [UPLOADED PYQ]	✓ Complete
	11	Regular Expressions (Pattern Matching & Parsing)	✓ Complete
	12	Language Models Overview (Sequence Probability)	✓ Complete
M2: LMs, POS & Parsing (12)	13	N-Gram Language Models (Unigram, Bigram, Trigram MLE) [UPLOADED PYQ]	✓ Complete
	14	Smoothing & Perplexity (Laplace Add-1 & Kneser-Ney) [UPLOADED PYQ]	✓ Complete
	15	Rule-Based POS Tagging (Handcrafted Morphological Rules)	✓ Complete
	16	HMM-Based POS Tagging & The Viterbi Algorithm [UPLOADED PYQ]	✓ Complete
	17	Context-Free Grammars (CFG Formal 4-Tuple & Parse Trees)	✓ Complete
	18	Recursive Descent Parsing (Top-Down Search)	✓ Complete
	19	Probabilistic Parsing (PCFG, CNF & CKY Algorithm) [UPLOADED PYQ]	✓ Complete
	20	Dependency Parsing (Typed Dependency Trees & Arc-Standard)	✓ Complete

Module	Topic #	Topic Name	Status
	21	Precision Metric Formulation in NLP Evaluation	✓ Complete
	22	Recall Metric Formulation in Extraction Tasks	✓ Complete
	23	F1-Score Harmonic Mean Formulation	✓ Complete
	24	Parsing Evaluation Metrics (UAS & LAS Attachment Scores)	✓ Complete
M3: Semantics & Vectors (11)	25	Lexical Semantics (Synonymy, Antonymy, Hyponymy, Meronymy)	✓ Complete
	26	Word Sense Disambiguation (Simplified Lesk & Supervised WSD)	✓ Complete
	27	WordNet Lexical Database (Synsets & Hypernym Hierarchies)	✓ Complete
	28	Lexical Databases & Structured Taxonomies	✓ Complete
	29	Semantic Similarity Measures (WordNet Shortest Path)	✓ Complete
	30	Cosine Similarity Formulation in Vector Spaces	✓ Complete
	31	Vector Space Models (Term-Document & Term-Term Matrices)	✓ Complete
	32	TF-IDF Weighting Formulation & Numerical Analysis [UPLOADED PYQ]	✓ Complete
	33	Word2Vec Embeddings (CBOW vs. Skip-Gram Architectures) [UPLOADED PYQ]	✓ Complete
	34	GloVe (Global Vectors from Word-Word Co-Occurrence)	✓ Complete
	35	FastText (Subword Character N-Gram Vector Representations)	✓ Complete
M4: Neural NLP (8)	36	Convolutional Neural Networks (CNNs) for Text Classification	✓ Complete
	37	Recurrent Neural Networks (RNNs) & Vanishing Gradients	✓ Complete
	38	Long Short-Term Memory (LSTM Gate Formulations: Forget, Input, Output)	✓ Complete
	39	The Transformer Architecture (Scaled Dot-Product Self-Attention)	✓ Complete
	40	Sequence-to-Sequence Encoder-Decoder with Cross-Attention	✓ Complete

Module	Topic #	Topic Name	Status
	41	Transfer Learning in NLP (Pre-Training + Task Fine-Tuning)	✓ Complete
	42	BERT (Bidirectional Encoder Representations from Transformers)	✓ Complete
	43	GPT (Generative Pre-Trained Causal Autoregressive Decoders)	✓ Complete
M5: Apps & Ethics (10)	44	Text Classification (Document Categorization & Naive Bayes)	✓ Complete
	45	Sentiment Analysis (Document, Sentence & Aspect-Based)	✓ Complete
	46	Named Entity Recognition (NER Sequence Labeling vs. POS) [UPLOADED PYQ]	✓ Complete
	47	Machine Translation Overview & The Vauquois Triangle	✓ Complete
	48	Rule-Based Machine Translation (Direct, Transfer & Interlingua)	✓ Complete
	49	Statistical Machine Translation (SMT Noisy Channel Formulation)	✓ Complete
	50	Neural Machine Translation (End-to-End Transformer Seq2Seq)	✓ Complete
	51	Chatbots & Conversational Dialogue Systems (NLU-DM-NLG)	✓ Complete
	52	Bias, Fairness & Stereotyping in Large Language Models	✓ Complete
	53	Model Explainability & Interpretability (Attention Heatmaps & LIME)	✓ Complete

★ Tier-S Top 16 University PYQ Exam Problem Types

#	PYQ Problem Type	Core Method / Mathematical Formula
1	Bigram MLE Probability	$P(w_i w_{i-1}) = \frac{C(w_{i-1}, w_i)}{C(w_{i-1})}$
2	Laplace Add-1 Smoothing	$P_{\text{Laplace}} = \frac{C+1}{C(w_{i-1})+ V }$
3	Language Model Perplexity	$PP(W) = P(W)^{-\frac{1}{N}}$
4	HMM Transition / Emission	$a_{ij} = \frac{C(t_i, t_j)}{C(t_i)}, b_j(o_k) = \frac{C(t_j, o_k)}{C(t_j)}$
5	Viterbi Decoding Algorithm	$V_i(j) = \max_i [V_{i-1}(i) \cdot a_{ij}] \cdot b_j(o_t)$
6	Stemming vs. Lemmatization	Rule-based chopping vs dictionary root morphological lemma
7	Top-Down vs. Bottom-Up Parsing	Recursive descent from $\bar{S}$ vs Shift-Reduce from terminal words
8	Chomsky Normal Form (CNF)	$A \rightarrow BC$ or $A \rightarrow a$ (Binary branching & terminal productions)
9	CKY Parsing Algorithm	$O(n^3 \cdot R)$ dynamic programming table parsing on CNF grammars
10	TF-IDF Term Weighting	$TF(t, d) \times \log_{10} \left(\frac{N}{DF(t)} \right)$
11	Word2Vec (CBOW vs. Skip-Gram)	Context $\rightarrow$ Target (CBOW) vs Target $\rightarrow$ Context (Skip-Gram)
12	WordNet Lesk Disambiguation	Overlapping dictionary definition glosses with surrounding context sentence
13	Transformer Self-Attention	$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{QK}^T}{\sqrt{d_k}} \right) \mathbf{V}$
14	BERT vs. GPT Architectures	Bidirectional Masked Encoder (BERT) vs Causal Autoregressive Decoder (GPT)
15	NER vs. POS Tagging	Semantic entity spans ('PERSON', 'ORG') vs syntactic grammatical word classes
16	SMT Noisy Channel Model	$\hat{T} = \arg \max_T P(S T)P(T)$ (Translation Model $\times$ Language Model)